Section A – Answer any 2 out of 3 questions in section A

Question 1

a) Why is the per-unit system used in analysis of power systems?

[4]

b) A network, composed of a transformer and transmission line, supplies demand as shown in Figure 1a. Transformer rated primary and secondary voltages and apparent power are denoted as V_{R1}, V_{R2}, S_R, respectively. Line parameters are length, L, resistivity r' (Ω/km), reactivity x' (Ω/km), and susceptibility b' (S/km). Assuming apparent power base, S_{base}, and adequate voltage base value, state or derive the equations for each of the per-unit terms in Figure 1b (p_G, q_G etc.) based on terms in Figure 1a.

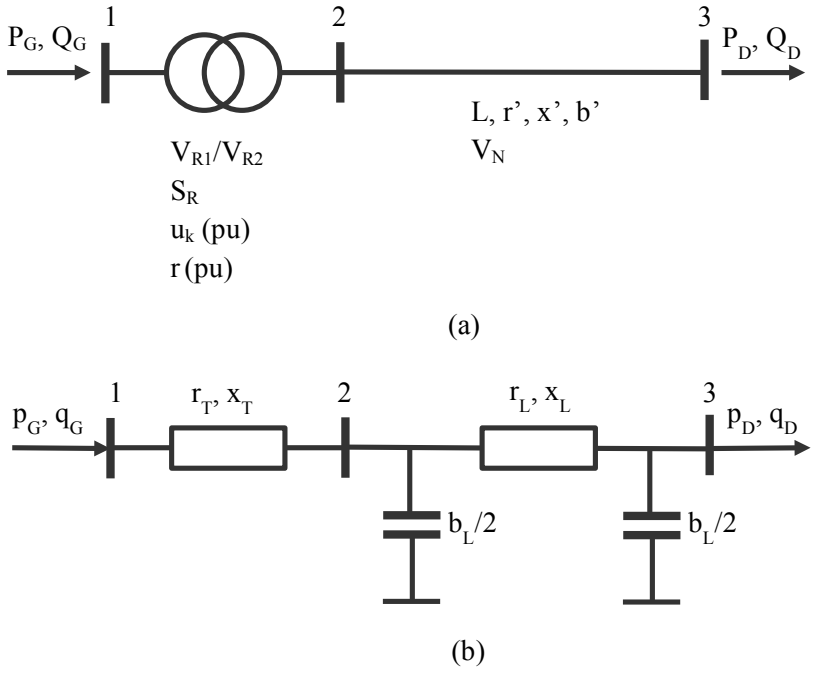


Figure 1: (a) Single line diagram of a network and (b) corresponding per unit representation

[8]

c) Assuming S_{base}=100 MVA, for the transformer and overhead line shown in Figure 1 calculate the per unit parameters if the transformer parameters are 400/136 kV, 240 MVA, u_k=0.2 pu, r=0.05 pu and the 132 kV line parameters are L=50 km, r'=0.17 Ω/km, x'=0.38 Ω/km, b'=7.5 μS/km.

[4]

d) A low voltage network supplies 100 households. Peak demand of each household is 10 kW and the coincidence factor for an infinite number of households is 0.1. In each household, a heat pump of 5 kW will be installed. If the coincidence factor of an infinite number of heat pumps is 0.6 calculate the new coincidence factor for an infinite number of households assuming that peaks of households and heat pumps coincide. Estimate the new household coincidence factor for an infinite number of households. Perform a calculation assuming 100 households. Briefly explain the benefits of diversification.

[4]

Question 2

a) For a synchronous generator, derive equations for the active and reactive power delivered to the system if voltages, angle and impedance are known

[3]

b) Consider a three-phase synchronous generator, 200 MVA, 21 kV, connected to a very large system. Synchronous reactance of the generator is $X = 1.5$ p.u. with negligible armature resistance. Terminal voltage $V = 21$ kV, and the generator is supplying the active power of 150 MW at 0.90 power-factor (reactive power is supplied to the system). Find:

(i) Magnitude and phase angle of the current. [3]

(ii) Magnitude and phase angle of the synchronous internal voltage. [3]

(iii) Per unit magnitude of the synchronous internal voltage. [2]

(iv) Magnitude and angle between internal (E) and terminal voltage (V) if reactive power delivered to the system is doubled.

[3]

c) Compute the probability that a system of 2 identical generators, each of 90 MW capacity and availability of 98%, and one wind farm of 30 MW and availability of 50%, will not meet peak demand of 100 MW.

[6]

Question 3

A three-bus power network is presented in Figure 3 below. Data relevant for the load flow analysis on this system are given in per unit values.

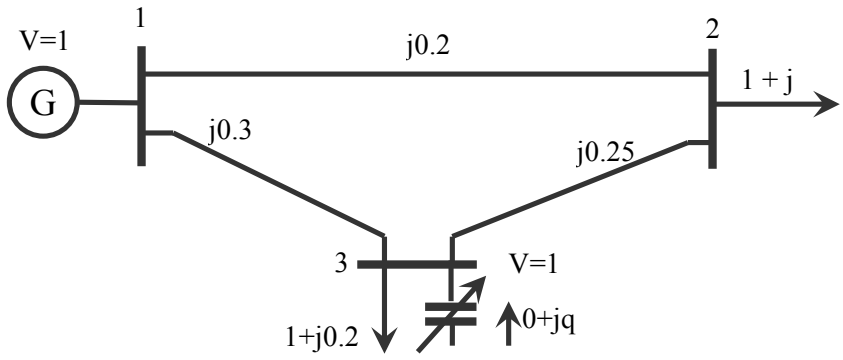


Figure 3. Three bus network showing per unit data for voltages,

active and reactive loads and line reactances

a) Explain the concept of the PV bus in the load flow calculations

[2]

b) Form the Ybus matrix and explain why it is singular (i.e. not invertible)

[2]

c) Perform 2 iterations of the Gauss-Seidel load flow method

[9]

d) Using the results from c), calculate the power mismatch at the PQ bus

[4]

e) Calculate the reactive power from compensator at the PV bus

[3]

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) Short circuit calculations may be simplified by neglecting the pre-fault currents and assuming the pre-fault voltages to be 1.0 p.u. Justify use of these assumptions in the context of high voltage bulk power transmission system.

[4]

b) What are the four steps for performing balanced fault current calculation without any simplifying assumptions about pre-fault voltages and currents?

[4]

c) Explain, using the theorem of constant flux linkage, why there is a decaying AC component in the stator current immediately after a three-phase short circuit on a generator terminal.

[5]

d) Two 100 MVA, 20 kV three-phase generators are connected to a 132 kV busbar through their individual generator transformer. The sub-transient reactance of both generators is 0.2 p.u. and the leakage reactance of both transformers is 0.1 p.u. From the 132 kV busbar, a 40 km long 132 kV line supplies an 11 kV load through a 200 MVA, 132/11 kV transformer with a leakage reactance of 0.1 p.u. The line impedance per phase is $(0.035 + j0.14)\ \Omega$/km. Using a 100 MVA base, calculate the following for a balanced three-phase short circuit on the low-voltage terminals of the load transformer:

(i) fault current (in kA) through the 132 kV line.

[5]

(ii) rating of the circuit breaker required at the load end of the 132 kV line.

[2]

Question 5

a) Starting from the fault current and voltage in phase domain explain how the three sequence networks should be connected for analysing the following unbalanced faults:

(i) line to line to ground (LLG) fault affecting phases B and C with a fault impedance Z_f to ground

[5]

(ii) line to ground (LG) fault on phase A with a fault impedance Z_f to ground

[5]

b) A 66 kV, three-phase transmission line is supplied by a 11 kV generator through a 11/66 kV transformer having star connection with grounded neutral on both primary and secondary sides. The positive-sequence reactance of the transformer and the transmission line are $j10.9\ \Omega$ and $j44\ \Omega$, respectively. The equivalent positive and negative-sequence sub-transient reactance of the generator, referred to the line voltage of 66 kV, are $j18\ \Omega$ and $j14.5\ \Omega$, respectively. The equivalent zero sequence reactance measured from the generator up to the fault point is $j150\ \Omega$ referred to the line voltage of 66 kV. For faults at the load end of the 66 kV transmission line, calculate the following neglecting the fault impedance and any resistance. Use 100 MVA base for the calculations.

(i) Fault current (in kA) due to a line to ground (LG) fault on phase A.

[5]

(ii) Current in phase B (in kA) during a line to line to ground (LLG) fault affecting phases B and C.

[5]

Question 6

a) Consider the standard power angle characteristic for a round rotor generator with zero resistance. Explain physically why the segment of the power angle characteristic with a positive slope corresponds to stable operation in steady state while that with a negative slope represents instability.

[4]

b) "The active power delivered by a round rotor synchronous generator depends on the angular separation between the rotor and the total magnetic field produced by the stator and the rotor windings". Justify this statement using a phasor diagram. Neglect any resistance of the generator.

[5]

c) A round-rotor synchronous generator with a synchronous reactance of 1.0 p.u. is connected to two parallel lines through a step-up transformer with 0.1 p.u. leakage reactance. The reactance of each line is 0.4 p.u. At the other end of the lines a step-down transformer with 0.1 p.u. leakage reactance supplies a synchronous motor on the LV side. The motor is of the round-rotor type and has a synchronous reactance of 1.0 p.u. The internal voltage of the generator is 1.2 p.u. and of the motor 1.0 p.u. If the generator produces 0.3 p.u. active power, determine the critical clearing angle for the following fault condition on one of the lines. Neglect the resistances and assume the power transfer between the generator and the motor to be zero during the fault.

(i) when the line fault is temporary and is cleared by itself.

[6]

(ii) when the line fault is permanent and is cleared by taking the faulty line out of service.

[5]